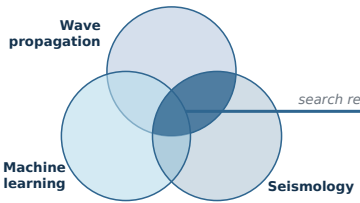
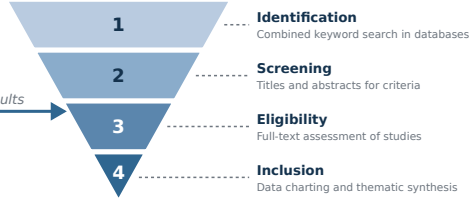
A Scoping Review of Physics-Informed Machine Learning for Wave Propagation Modeling in Seismology

1 Literature search



2 Scoping review steps



Identification
Combined keyword search in databases

Screening
Titles and abstracts for criteria

Eligibility
Full-text assessment of studies

Inclusion
Data charting and thematic synthesis

3 Trends



Frequent approaches in seismology

Fourier neural operator	Generative adversarial
Recurrent network	Dual networks

4 Challenges and oportunities

Comparison
Accuracy and efficiency must be benchmaked against standard methods

Evaluation
Must consider the full workflow: data generation, training, evaluation, and reuse across problems

Scalability
3D, high-frequency and field-scale validation remains an open challenge